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Panel data estimates of the effects of different types of crime on housing prices

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ABSTRACT

One of the most studied effects of crime is the impact that neighborhood crime has on housing values. A major drawback of these studies is that, although crime is undoubtedly endogenous in property value models because of either simultaneity, omitted variables or measurement error, the vast majority of studies treat crime measures as exogenous independent variables. We exploit a unique nine-year crime panel at the neighborhood level to estimate models that properly address the endogeneity of crime and allow us to overcome other specification errors that have plagued previous studies. Of the seven different types of crime we investigate, only robbery and aggravated assault crimes (per acre) exert a meaningful influence upon neighborhood housing values.

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1. Introduction

Maintaining public safety is a major responsibility of local governments that accounts for a significant percentage of their budgets.[1] To spend these dollars most effectively, reliable estimates of the costs of crime are needed to guide policy decisions. Unfortunately, obtaining estimates of these costs has been proven to be a difficult task, principally because public safety (i.e., the absence of crime) is a non-market good, whose price can only be estimated implicitly. Typically, the market chosen to estimate the value of crime prevention implicitly is the housing market, where a hedonic price model is estimated that includes a measure of neighborhood crime among the regressors.

The hedonic model approach to estimating the costs of crime is theoretically attractive. The seriousness that people attach to crime should be reflected in what they are willing to pay to live in a low crime neighborhood. Moreover, by including crimes of different types in the hedonic model, their relative importance can be determined. However, a major difficulty arises in estimating the crime–housing price relationship; namely, the endogeneity of crime must be addressed in order to obtain consistent estimates. While the endogeneity of crime is widely recognized, it is remarkable that of the 19 hedonic studies we reviewed in earlier work (Ihlanfeldt and Mayock, forthcoming), 13 treated crime as an *exogenous* variable. Of the six studies that do instrument crime, only two tested the validity of the instruments, and in one case an overidentification test *rejected* the exogeneity of the instruments. Undoubtedly, the endogeneity of crime has been skirted in the literature for but one reason — it is extremely difficult to identify variables that satisfy the conditions required of a valid instrument: 1) that the variables be highly correlated with the endogenous crime measure, and 2) that the variables be uncorrelated with the error term and correctly excluded from the estimated housing price equation.

We exploit a unique nine-year crime panel at the neighborhood level to estimate models that properly address the endogeneity of crime and allow us to overcome other specification errors that have plagued previous studies. Our neighborhoods are aggregations of one-mile square areas that are combined by design to approximate census tracts. The latter are the areas most frequently used in prior crime studies to represent neighborhoods. Besides approximating areas consistent with those employed in prior work, our neighborhood units are sufficiently large to contain enough single-family home repeat sales to estimate separate repeat sales price indexes for each neighborhood. Because we have the geographical coordinates of each crime, we aggregate them to the level of our neighborhoods. Our crime data and house price data come from Miami-Dade County, Florida, and they span the years 1999–2007. The endogeneity of crime is addressed in two ways — we first difference the housing price index and the crime measures which controls for various endogeneities in crime levels, and we instrument changes in crime to allow for correlation between the crime measures and current and past values of the idiosyncratic error. This first-differences instrumental variables estimator allows us to consistently estimate the implicit costs of

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[1] Expenditure data from the Florida Department of Financial Services indicates that in 2006, Miami-Dade County, the subject of our empirical investigation, spent over $540 million on law enforcement. This figure constituted over 15% of all county expenditures in this fiscal year.

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various crimes: murder, robbery, aggravated assault, burglary, motor theft, larceny, and vandalism.[2] Our results indicate that only an increase in the density of aggravated assault and robbery crime lower neighborhood housing values. A 1% increase in our preferred measure of neighborhood crime (number of crimes per acre) is found to reduce housing price by between .1 and .3%.[3]

2. The endogeneity of crime and previous studies

There are at least five mechanisms whereby crime may be endogenous in a housing price model. Three result in a positive crime-housing price relationship, while the other two cause crime and price to be negatively related. Hence, the failure to account for the endogeneity of crime may either upwardly or downwardly bias estimates of the effect of crime on property value, depending on the relative strengths of these various mechanisms.

One of the mechanisms in the first group is that neighborhoods with more expensive homes attract criminals by offering higher expected payoffs in terms of the market value of stolen goods. Another of these mechanisms is that crime statistics are limited to only those crimes that are reported to police. Reporting rates are known to be higher in more affluent neighborhoods (Skogan, 1999). Finally, some unobservables that increase the attractiveness of a property (e.g., large windows or a secluded back yard) also make the property an easier target for crime (Gibbons, 2004).

In the second group of mechanisms (i.e., those resulting in a negative crime–housing price relationship) is the relationship resulting from the deterrence provided by self-protection measures. Self-protection is expected to be greater in wealthier neighborhoods because property owners are more able to afford it and they have more at risk. The other mechanism in this group is that neighborhoods with cheaper housing attract lower-income individuals, and it is well known that income and the propensity to commit crime are inversely related.[4] Evidence also exists which shows that criminals commit the vast majority of their crimes within their home neighborhood (Reppetto, 1974; Pope, 1980). The implication is that criminals self-select into neighborhoods with lower property values and commit many of their crimes within these neighborhoods.

As noted above, while most studies have treated crime as exogenous to housing price, six studies have treated crime as an endogenous variable.[5] However, only one of these studies (Gibbons, 2004) fully validates the choice of instrumental variables (both first-stage regression results and, where equations are overidentified, tests of overidentifying restrictions are reported); and the instruments used by *all* studies, with the possible exception of Gibbons', are open to question.[6] We therefore only discuss Gibbons' study below, but the details of all six studies, including findings, are summarized in Table 1.[7]

Using 1999–2001 data for London, England, Gibbons (2004) regresses individual house prices on burglary and vandalism crimes per square kilometer, after assigning both homes and crimes to a 100 meter grid. In the fashion of a standard fixed effects estimator, Gibbons first transforms his data by taking deviations from local district averages of the variables. In this way, unobservables that are constant within the district are eliminated as a source of endogeneity. To further address the endogeneity of crime he instruments his two crime variables (the density of burglary and vandalism crimes against residences) using as instrumental variables the density of burglary and vandalism crimes against non-residential buildings. In an alternative model, distance to the nearest public house or wine bar, and its polynomials, are used to instrument the density of vandalism crimes, based on the argument that alcohol consumption is an associated factor in many types of crimes. The instruments are found to be significantly correlated with the crime densities and pass Hansen's overidentification test.

Gibbons' results indicate that vandalism but not burglary crimes have an impact on home prices. He explains his somewhat counterintuitive results by arguing that acts of vandalism (but not burglaries) "motivate fear of crime in the community and may be taken as signals or symptoms of community instability and neighborhood deterioration in general" (Gibbons, 2004: 441). Because vandalism crimes are included in our data, we are able to further investigate the importance of this category of crime relative to, not only burglary, but all other major categories of crime.

Apart from the endogeneity of crime problem, another significant limitation of previous studies concerns the chosen measure of crime. All studies (including those that endogenize crime) use either total crime or a single type (in some cases a couple of types) of crime as the crime measure. The former studies simply count up the number of crimes, regardless of their nature, which implicitly gives each type of crime the same weight. Generally, total crimes are limited to those tracked by the U.S. Federal Bureau of Investigation and labeled by the FBI as 'Part I Crimes' or 'indexed crimes.' These crimes include homicide, aggravated assault, rape, robbery, burglary, larceny, and auto theft. Clearly these crimes differ greatly in their severity, and there is no reason, *a priori*, to believe that a rape or murder should be equated with a larceny or auto theft.[8] Hence, using a single total crimes measure is subject to considerable measurement error. Measuring just a single type of crime (or a few types) is not a solution to this problem because, as we document below, all indexed crimes are highly collinear. Neighborhoods with a violent crime problem are also those with a property crime problem. Hence, substituting a type of crime for total crime as the crime measure trades a measurement error for an omitted variable problem. Moreover, this correlation between different types of crime also invalidates the approach of estimating a sequence of price models in which a different type of crime is included in each equation.

In light of these criticisms, to avoid obtaining biased estimates, the preferred approach is to include measures of the incidence of each type of crime in the hedonic model. The problem, of course, as alluded to above, is that different types of crimes are highly collinear; hence, multicollinearity makes it difficult to separate out their separate influences. One solution to this problem is to have panel data and first difference the data. As we document below, crimes by type are far less collinear in changes than in levels.[9]

[2] The first six crimes are six of the seven "indexed" crimes identified by the FBI. The excluded indexed crime is rape. To protect the identity of victims of rape, the addresses of rape crimes are not reported and therefore could not be assigned to census tracts.

[3] As we argue below, at the neighborhood level, crimes per acre (crime density) has more intuitive appeal than crimes per person (crime rate). We also will show that the former crime measure has the greater explanatory power.

[4] For a review of the literature that links income to crime, see Eide (1994).

[5] For a recent review of the crime-housing price literature, see Ihlanfeldt and Mayock (forthcoming).

[6] Apart from the endogeneity of crime problem, as we discuss below, there are other possible specification errors that may have biased the results of earlier studies.

[7] Our focus on Gibbons' study also stems from our desire to further investigate his finding that vandalism has an important effect on property value. This crime category has seldom been analyzed in previous studies.

[8] A number of studies have attempted to calculate the cost incurred by victims of different types of crimes, where total cost is inclusive of both the out of pocket (monetary) cost and the psychic cost associated with pain and suffering. All of these studies demonstrate that violent crime carries a higher cost than property crime. For example, Cohen et al. (1995) estimate that the total cost of a rape is $87,000, while that of a motor vehicle theft is $4000 (measured in 1993 dollars).

[9] Omitted variables bias may also result from excluding non-crime variables that are correlated with both neighborhood crime and neighborhood housing prices, such as the presence of vacant or abandoned housing in the neighborhood. To the extent that the variance in these variables is across rather than within neighborhoods, first differencing the data also controls for possible omitted variables bias. In addition, as discussed below, we add to our models variables that may capture changes in neighborhoods over time that may also mitigate possible omitted variables bias.

Table 1
Crime/property value studies that have endogenized crime.

Author(s) and date of publication	City	Year(s)	Data	Crime variables	Instruments	Validation of instruments	Findings
Rizzo (1979)	Chicago	1968–1972	1970 median rent for 111 blocks; 1970 median owner-estimated value for 68 blocks; 324 single-family sales; for above three data sets total crime, violent crime, property crime by block; median owner-estimated house value for 71 neighborhood communities; for the latter data, the 71 communities are assigned total crimes reported for 21 police districts	For block and individual sales regressions: total crime rate, property crime rate, violent crime rate of block; for 71 neighborhood communities, total crime rate of police district	Crime is instrumented in only the regression run for the 71 neighborhood communities, instruments = the proportion of population 15–24, median years of schooling, unemployment rate, population density, proportion of population on welfare, males to females, and labor force participation rate	None	Generally, all crime variables have negative and significant effects across the alternative databases. For the database where crime is instrumented, estimated crime coefficients are much larger than those obtained with OLS.
Naroff et al. (1980)	Boston	1970	Median owner-estimated house value for census tracts; total crimes of tract	Crime rate of tract	Population density of tract	None	Elasticity of house value with respect to crime rate = −1.67.
Burnell (1988)	Chicago suburban communities	1980	Median owner-estimated value for 71 communities; property crime for each community	Property crime rate	Community fiscal and demographic characteristics	None	Elasticity of house value with respect to property crime rate = −.10.
Buck et al. (1993)	Atlantic City region	1972–1986	Total assessed value for 64 communities; larceny, auto theft, burglary crimes for each community	Larceny, auto theft, burglary crime rate of community	Lags of explanatory variables	Failed overidentification test	Highly mixed, with no consistent finding across alternative model specifications.
Gibbons (2004)	London	1999–2001	10,464 individual single-family sales; burglaries, vandalism crimes for the 100 meter grid	Burglaries/kilometers, vandalism/kilometers	Commercial crime densities, distance to nearest public house or wine bar	Yes	A one-tenth standard deviation decrease in the local density of vandalism crime adds 1% to house price; burglary crime rate not significant.
Tita et al. (2006)	Columbus	1995–1998	43,000 single-family sales; index crimes by census tract	Total crime rate, violent crime rate, property crime rate (entered in levels and changes); separate regressions run for each type of crime	Murder rate	None	Crime is capitalized into house price at different rates for poor, middle, and wealthy neighborhoods, with violent crime most important.

Having provided a critical review of the extant literature that has treated crime as an endogenous variable in a house value (or rent) model, we next provide some new evidence that addresses the limitations that we have identified in previous studies.

3. Data

Our panel data set is constructed from two sources — the Miami-Dade County property tax rolls obtained from the Florida Department of Revenue (FDOR) and a database of crime incident reports maintained by the Miami-Dade County Sheriff's Office. These data sources and how we used them to construct our panel are described below.

Florida statutes require that each county annually submits its property tax roll in a standardized format to FDOR for auditing purposes. These rolls span the years 1999 to 2007.[10] Among the items included on the rolls is a detailed code indicating the current land use of each parcel, the interior usable space for improved properties, and the price and date of sale for the two most recent property transfers. All transfers are included in the data, both commercial and residential, but we use only single-family home sales in constructing our neighborhood price indexes.

Our goal is to investigate the impact of crime on property values at the neighborhood level. To accomplish this we seek to place crimes and house sales within neighborhoods for all of the years included in our panel. Helpful in this regard is that we have the geographical coordinates of each crime and a digitized parcel identification map for 2007, the final year of our panel. Unfortunately, we do not have these parcel maps for previous years. One possible solution to this problem is to use the 2007 map to identify the locations of sales that occurred prior to 2007. The parcel identification number of many of these sales, however, either cannot be found on the 2007 map or the parcel identification number from the previous tax rolls identifies the wrong parcel on the 2007 parcel map. Another solution is to use the township, range, and section (TRS) identifiers found on each tax roll to place each sale and crime within a one-square-mile area.[11] These TRS squares do not change over time, so that they provide the building blocks from which our neighborhoods can be constructed.

The neighborhood area in almost all previous studies of the crime/housing price relationship has been the census tract, which is the U.S. Bureau of the Census' geographical area that best represents a neighborhood. Hence, a reasonable choice would be to combine the TRS squares to approximate as closely as possible the census tract geography. Tract sizes in the unincorporated area of Miami-Dade County are sufficiently large relative to the TRS squares that our neighborhood boundaries match up fairly well with those of census tracts.[12]

Having defined our neighborhoods, we then construct a number of variables to be used in our empirical model. One set of variables captures inter-neighborhood and intertemporal differences in commercial land use composition. More specifically, for each of 26 different commercial land use classifications defined on the FDOR tax rolls, we construct a continuous variable measuring the total amount of interior square footage of that land use type within the neighborhood. It is from these commercial land use variables that we derive our instrumental variables, as described below.

Another variable we construct for each neighborhood/year record is a constant-quality house price index value. This value is obtained by using the information on the two most recent sales transactions of each single-family home to estimate standard repeat sales models separately for each neighborhood using the following model:

$$\ln\left(\frac{P_{i,t}}{P_{i,t-n}}\right) = \sum_{k=1}^{T} \beta_k D_{i,k} + \varepsilon_{i,t,t-n} \quad (1)$$

where $P_{i,t}$ is the most recent selling price of property i at time t;
$P_{i,t-n}$ is the previous selling price of property i at time $t-n$;
β_k is the regression coefficient on $D_{i,k}$;
T is the last time period in the sample;
k is an index over the time periods;
$D_{i,k}$ is a dummy variable which equals -1 at the time of the initial sale, $+1$ at the time of the second sale, and 0 otherwise; and
$\varepsilon_{i,t,t-n}$ is the regression error term.

If β_t denotes the estimate of the year t coefficient, the price index value for year t (I_t) equals $\exp(\beta_t)*100$.

Besides the FDOR tax rolls, our other source of data are the crime incident reports provided by the Miami-Dade County Sheriff's Office. This database, which contains information on more than 2 million crimes occurring between 1999 and 2007, identifies the date, time, and type (e.g., burglary and robbery) of each crime reported in unincorporated Miami-Dade County and the municipalities of Miami Gardens, Miami Lakes, Doral, Palmetto Bay and Cutler Bay. As of 2007, this collection of six jurisdictions had a total population of 1.3 million people, which amounts to roughly 54% of the total population of Miami-Dade County. The most unique aspect of the crime database is that it contains the exact location where the crime was committed. This enables us to assign crimes to neighborhoods in the same way we assigned home sales to neighborhoods, as described above. Crimes are broken down into seven major categories: homicide, aggravated assault, robbery, burglary, motor theft, larceny and vandalism. The first six types of crime are six of the seven major crime types identified by the FBI. We add to this list vandalism, based on Gibbons (2004). Recall that he found that vandalism has a strong negative effect on property value, presumably because it signals to home buyers future neighborhood decline.

In the literature that has investigated the effect of crime on property values, crime has been measured at the neighborhood level in three different ways: number of crimes vs. number of crimes per resident vs. number of crimes per acre (or other spatial unit). The latter two measures are commonly referred to as the crime rate and the crime density, respectively. Of the three alternative measures, the crime rate has been the most popular measure, presumably because it best registers the individual resident's risk of victimization. Criminologists have argued, however, that at the neighborhood level (in comparison to the city level, which is the level at which crime rates are reported by the FBI) the crime rate is a less reliable measure of a resident's probability of being victimized because crime is higher where business activity is greater and the latter activity varies greatly across census tracts.[13] Moreover, Bowes and Ihlanfeldt (2001) have argued that crime density may be more highly correlated with the average resident's knowledge of crimes committed in his neighborhood. Crime density may also be a better measure of neighborhood crime than the crime rate if people wish to avoid exposure to crime. The latter equals the probability of being either a victim of crime or a

[10] As is well known, in recent years Florida has seen its real estate bubble burst, so there may be concerns surrounding our use of data containing a major turning point. However, according to the repeat-sales price index published by the Office of Federal Housing Enterprise Oversight for Miami-Dade County, the housing price reversal did not begin in Miami-Dade County until the third quarter of 2007, which represents the very end of our panel.

[11] The TRS identifiers come from the Public Land Survey System (PLSS). More information on the PLSS can be found at http://nationalatlas.gov/articles/boundaries/a_plss.html. Each township/range/section combination typically corresponds to a one-mile by one-mile square. Although quite rare, it can be the case that the TRS code identifies a geographic area that is larger or smaller than 1 mile^2. This might be the case, for instance, in sections near jagged borders (e.g., near the coast).

[12] A map with our neighborhood geography overlaid on the census tract geography is available upon request.

[13] See Harries (1981) for a review of the literature on this issue.

witness to a crime. This exposure probability is better measured by the number of crimes in a unit of space than by the number of crimes per resident. All three alternative crime measures are included in our panel so that we can experiment with each one in estimating our models.

Our panel includes nine years of data for 130 tracts, yielding a total of 1170 tract/year observations. As described below, we first difference the data and allow for distributed lag effects going back four years. This leaves us with 520 observations. For 18 of these observations a house price index value could not be computed due to an insufficient number of repeat sales transactions, resulting in a final sample size of 502 observations.[14]

4. Models used to estimate the housing price–crime relationship

To estimate the effect of crime on housing price, we first difference the data and regress the change in the housing price index on changes in crime. Focusing on housing price *changes* rather than *levels* helps mitigate possible bias resulting from each of the sources of the endogeneity of crime identified above. For example, crime may be worse in neighborhoods with lower housing prices because criminals are more likely to live in these neighborhoods and, as noted above, criminals tend to commit their crimes close to where they live. Criminals may also be more likely to live in neighborhoods experiencing less price appreciation. To the extent that this relationship between the location of criminals and neighborhood affluence is time-invariant, the standard first-differences estimator provides consistent estimates of the effect of crime on housing prices in the presence of arbitrary correlation between the regressors and that portion of the composite error term that is constant within the tract.[15] Similar arguments can be made in favor of first differencing the data for the other four sources of crime endogeneity.

Another advantage from first differencing the data is that multicollinearity among the variables measuring each type of crime is substantially reduced. Table 2 presents correlation coefficients, in both levels and changes, using as the crime measure the number of crimes per acre within the neighborhood.[16] Levels of crime density are highly collinear; for example, the correlation coefficient between the densities of robbery crime and aggravated assault crime is .91. In contrast, the correlations between changes in crime densities are all low, with the highest being .24 between the changes in robbery and motor theft density.

The final advantage from first differencing the data is that it controls for variables that may affect both crime and housing prices that are stable over time. Hence, the possibility of omitted variables bias is mitigated.

The basic model we estimate to investigate the effect of crime on housing price can be expressed as:

$$\Delta I_{i,t} = \gamma_t + \sum_{j=1}^{J} \sum_{h=0}^{H} \beta_{j,t-h} \Delta C_{i,j,t-h} + \Delta \varepsilon_{i,t} \quad (2)$$

where ΔI_{it} is the change in the repeat sales price index in tract i between year t and year $t-1$; γ_t are time fixed effects (i.e., year dummy variables); and j represents the seven different types of crime.

For each type of crime, we include the current change $(C_t - C_{t-1})$ and four lags $(C_{t-1} - C_{t-2}, C_{t-2} - C_{t-3}, C_{t-3} - C_{t-4}, C_{t-4} - C_{t-5})$. The expectation is that the full impact of a spike in crime on housing price will occur with a lag. First, it takes time for new information regarding the safety of a neighborhood to be fully assimilated by market participants. Second, current changes in housing prices reflect to some extent prices agreed upon before the increase in crime occurred, given that the price is recorded at the time of closing and considerable time may expire between when an agreement is reached and the date of closing.

The two most widely-used panel data estimators, the first difference estimator and the fixed effects estimator, are consistent under the assumption of strict exogeneity, which rules out any correlation between the regressors and the error term *in any time period, current, future or past* (Wooldridge, 2002: 254). An important implication of strict exogeneity is that current values of the dependent variable cannot affect current or future values of the explanatory variables. For example, if an increase in ΔI_t causes an increase in ΔC_t or ΔC_{t+1}, strict exogeneity is violated. Thus, although the first difference estimator is consistent in the presence of correlation between reported criminal activity and the neighborhood-specific fixed effect, the first difference estimator will generally be inconsistent if our crime variables are correlated with that portion of the composite error term that varies over time.

If strict exogeneity is violated, consistent estimates may be obtained by instrumenting ΔC_t and possibly ΔC_{t-1}, depending on one's assumptions. If we assume that there is no contemporaneous feedback and ΔI_t is allowed to affect only future changes in crime $(\Delta C_{t+1}, \Delta C_{t+2}, \ldots)$, then only ΔC_t requires instrumentation. However, if we allow for the crime levels to be correlated with contemporaneous as well as past values of the error term, then both ΔC_t and ΔC_{t-1} must be instrumented. It is reasonable to assume that there is no contemporaneous feedback, because it seems unlikely that criminals would react to changing neighborhood housing values without some delay. Just like home buyers in their response to changing neighborhood crime levels, it takes time for criminals to obtain and assimilate new information regarding neighborhood crime opportunities and change their behavior accordingly. Nevertheless, we estimate Eq. (2) instrumenting just ΔC_t, but also run regressions where both ΔC_t and ΔC_{t-1} are instrumented.

In most cases, when using panel data the choice of instruments is an easy one. Lagged levels of the endogenous change variables typically make good instruments because levels and changes are correlated and the lagged values are generally assumed to be exogenous. But this strategy will not work in our case, because lagged changes in crime enter our model as explanatory variables.[17] Another challenge that instrumentation poses for our model is that there are 14 variables (seven crime types, with ΔC_t and ΔC_{t-1} endogenous) that must be instrumented, if we allow for the possibility of contemporaneous correlation between crime levels and the error term.

Our instrumentation strategy involves using our data on the commercial land uses that exist within each neighborhood. Recall that there are 26 different commercial land use categories, and for each

[14] When estimating repeat sales indexes at the neighborhood level there is always the concern that too few repeat sales may be available to obtain precise estimates. We defined 30 as a sufficient number of repeat sales and dropped those neighborhoods with fewer than this number of sales. We also tried dropping those neighborhoods with less than 60 repeat sales, which reduced our sample from 502 to 461 observations. The estimated effects of changes in crimes on changes in the index using the 60-sales cut-off are nearly identical to those reported below using the 30-sales cut-off.

[15] After first differencing the data, the endogeneity issue is whether criminals are more likely to move into neighborhoods experiencing less price appreciation. This is not clear, but what is clear is that endogeneity arising from the movement of criminals is less likely to be a concern in comparison to that which arises from the location of criminals.

[16] As noted above, we use three alternative measures of crime — the number of crimes in the tract, the crime rate in the tract (crimes/population), and the crime density in the tract (crimes/acre). The latter variable is our preferred measure for reasons outlined in Section 3.

[17] For a discussion on instrumentation in distributed lag models, see Wooldridge (2002: 307).

Table 2
Correlation coefficients.

	Larceny	Burglary	Motor theft	Murder	Assault	Robbery	Vandalism	Δ Larceny	Δ Burglary	Δ Motor Theft	Δ Murder	Δ Assault	Δ Robbery
Larceny													
Burglary	.64												
Motor theft	.75	.81											
Murder	.36	.52	.40										
Assault	.53	.78	.66	.69									
Robbery	.63	.80	.74	.67	.91								
Vandalism	.73	.80	.78	.50	.74	.74							
Δ Larceny	−.02	−.02	−.08	−.02	−.09	−.06	−.06						
Δ Burglary	−.09	.10	−.13	−.03	−.15	−.10	−.10	.18					
Δ Motor Theft	−.09	−.04	.09	−.03	−.02	.01	−.06	.09	.01				
Δ Murder	.01	.02	−.03	.53	.04	.04	.03	.04	.04	−.13			
Δ Assault	−.12	−.14	−.11	−.13	−.15	−.21	−.17	.05	.12	.09	.06		
Δ Robbery	−.05	−.03	−.01	−.10	−.13	.05	−.04	.14	.16	.24	−.05	.12	
Δ Vandalism	.05	.06	.08	.01	.01	.02	.23	.10	.01	.08	.01	.07	.07

one we know the total square footage of interior space within the neighborhood. We first difference these data to obtain our instruments.

The logic underlying our choice of instruments is that the criminal's opportunity set depends on the commercial land uses that exist within the neighborhood. Some land uses are excellent targets for some types of crime, while other land uses favor other types of crime. For example, a new convenience store provides an opportunity to commit a robbery (a violent crime), while more offices create more locations that can be burglarized (a property crime). Hence, we expect that changes in particular types of crime are correlated with past changes in particular commercial land uses.

A concern regarding our instruments is that the commercial activity within a neighborhood may have positive or negative direct effects on nearby property values. Negative effects are possible if some types of commercial land use generate traffic, noise or other negative externalities. Positive effects are possible if commercial activity provides convenience for shopping or entertainment. However, these effects, whether positive or negative, are likely to be the result of differences in average levels of overall commercial activity rather than marginal changes within individual land use categories. For example, homes may sell for more in neighborhoods that offer more total shopping opportunities, but another shoe store is not likely to alter the attractiveness of the neighborhood sufficiently to affect property values.

In short, our argument is that a marginal change in a single commercial land use category will affect the opportunity landscape for criminal activity but will not affect the attractiveness of the neighborhood (apart from the effect that the change in land use has on attractiveness through its effect on crime).[18] One way to bolster our argument is to purge from our set of instruments those land use categories that include shopping, eating and drinking establishments. If marginal changes do influence property values, they would seem to matter more for these particular land uses, resulting in important changes in our results. As reported below, paring down our list of instruments in this way has little effect on our results.

Ultimately, the validity of our instrumentation strategy rests upon the performance of the instruments. Specifically, are they highly correlated with the endogenous variables? Are they correctly excluded from the price equation? Are they uncorrelated with the error term of the price equation? As reported below, statistical tests directed at answering these questions support our chosen instruments.

5. Spatial variation in neighborhood crime

Before discussing the results from estimating the house price models, the substantial variation in crime across our neighborhoods is shown with the data. First, the average number of crimes per acre and the average number of crimes per 1000 persons were computed for each neighborhood over the nine years that make up our panel. This statistic can be interpreted as the average reported criminal activity in the neighborhood over the duration of the panel.[19] Next, the mean of these averages was computed for each quintile in the crime distribution, where the bottom quintile is the 20% of neighborhoods with the lowest crime densities (rates) and the top quintile is the 20% of neighborhoods with the highest crime densities (rates). These means are reported in Table 3 for total indexed crime, violent crime (murder, robbery, and aggravated assault), property crime (burglary, motor theft, and larceny) and each of the seven individual types of crime.[20]

The mean number of total indexed crimes per acre (per 1000 population) is 13 (6) times greater in quintile 5 than in quintile 1. Substantial variation in crime is also found for each of the two major categories of crime. This is especially true for violent crime. For example, the average number of violent crimes per acre in the fifth quintile is 32 times greater than the density of violent crimes in the first quintile. Turning to the individual crimes, murder shows the greatest variation (density (rate) is 231 (94) times greater in quintile 5 than in quintile 1), while vandalism shows the least variation (density (rate) is 11 (5) times greater in quintile 5 than in quintile 1).

We also computed the average annual change in the crime density and in the crime rate over the panel for each neighborhood. The

[18] In the context of the economic theory of the supply of crime (Ehrlich, 1996), our identification strategy makes use of variables that shift the supply of criminal offenses but do not otherwise influence the value of housing. According to this theory: "A person's decision to participate in illegal activity can be viewed as motivated by the costs and gains from such activity" (Ehrlich, 1996). If the travel between a criminal's location and his target is costly, the rational criminal's decision calculus implicitly defines a distance beyond which the expected costs of crime outweigh the expected benefits of crime. Our argument is that changes in the land use mix at the neighborhood level influence the supply of crime by altering the direct costs associated with executing the crime. Consider our example of the addition of a convenient store, a popular target for robberies, to a neighborhood. For many criminals within this neighborhood, the convenient store will now be close enough to justify incurring the costs associated with a robbery, and criminal activity would increase within the neighborhood.

[19] It is important to note that our crime rate estimates are subject to two sources of measurement error because the denominator (i.e., the population of the neighborhood) used to construct the rate is obtained from census data. First, to obtain an estimate of the population of the neighborhood the proportion of land area falling within different census tracts is established from our GIS maps. These proportions are then used to estimate neighborhood population as a weighted average of tract populations. Second, we only have an estimate of neighborhood population for the year 2000. Hence, estimated annual changes in the crime rate come only from changes in the number of crimes reported for each year.

[20] Recall that we exclude one of the seven indexed crimes (namely, rape) because the location of these crimes is not reported.

Table 3
Neighborhood characteristics (means of 9-year averages).

	Mean	Means by quintile				
		(1)	(2)	(3)	(4)	(5)
Indexed crime[a]						
Density	.471	.076	.276	.414	.606	.964
Δ Density	−.013	−.048	−.019	−.007	.000	.008
Rate	67.6	23.7	40.9	54.7	79.6	138.1
Δ Rate	−1.5	−6.4	−2.1	−.8	.1	1.9
Violent crime[b]						
Density	.064	.000	.018	.034	.066	.194
Δ Density	−.002	−.008	−.002	−.000	.000	.002
Rate	9.0	1.6	2.9	4.5	8.9	26.1
Δ Rate	−.2	−1.1	−.2	−.1	.0	.3
Property crime[c]						
Density	.406	.070	.253	.376	.533	.788
Δ Density	−.011	−.041	−.018	−.006	−.000	.008
Rate	58.6	21.4	36.5	48.4	68.7	115.6
Δ Rate	−1.3	−5.6	−2.0	−.8	.2	1.9
Murder						
Density	.00072	.00001	.00014	.00028	.00064	.00231
Δ Density	.00003	−.00022	.00000	.00000	.00000	.00031
Rate	.10449	.00355	.02454	.05051	.09573	.33381
Δ Rate	.00274	−.03149	.00000	.00000	.00000	.04052
Robbery						
Density	.02350	.00167	.00543	.01105	.02384	.07347
Δ Density	−.00020	−.00299	−.00043	−.00003	.00035	.00194
Rate	3.19044	.43565	.86486	1.44887	3.05997	9.79920
Δ Rate	−.02000	−.39413	−.06357	−.00908	.05930	.28604
Aggravated assault						
Density	.04084	.00382	.01168	.02179	.04150	.12129
Δ Density	−.00171	−.00716	−.00154	−.00074	−.00005	.00095
Rate	5.69287	1.06026	1.87026	3.05704	5.67601	16.24779
Δ Rate	−.21438	−.96986	−.21054	−.08574	−.00415	.16275
Burglary						
Density	.07651	.01297	.04344	.06854	.09532	.15875
Δ Density	−.00202	−.00937	−.00266	−.00090	.00021	.00274
Rate	11.02576	3.56154	6.55013	9.30869	12.86757	22.12680
Δ Rate	−.18578	−1.17799	−.31086	−.12947	.05659	.61150
Motor theft						
Density	.06215	.00864	.03421	.05604	.08027	.13077
Δ Density	−.00325	−.00937	−.00481	−.00248	−.00063	.00107
Rate	8.54234	2.52785	5.28868	7.28303	10.14705	17.01793
Δ Rate	−.40412	−1.23127	−.55766	−.31566	−.10571	.20154
Larceny						
Density	.26807	.04733	.16583	.24547	.35401	.52771
Δ Density	−.00649	−.03057	−.01138	−.00260	.00073	.00836
Rate	39.07015	13.30703	23.43889	31.69156	44.71554	81.88034
Δ Rate	−.74229	−4.20786	−1.20427	−.36862	.18189	1.71297
Vandalism						
Density	.06071	.01067	.03961	.05979	.07543	.11425
Δ Density	.00119	−.00247	.00006	.00099	.00218	.00501
Rate	8.57333	3.04195	5.76920	7.77716	10.44967	15.64651
Δ Rate	.19901	−.33227	.01508	.16455	.37649	.76546
Household income	53,288	35,360	45,027	51,587	57,742	75,433
%Δ Index 2004–07	37.2	23.0	32.1	36.4	42.4	51.7
For assault crime[d]						
Household income		64,205	59,597	54,882	49,750	40,169
%Δ Index 2004–07		31.4	30.6	36.7	39.2	46.0

[a] Indexed crime = murder + aggravated assault + robbery + burglary + motor theft + larceny.
[b] Violent crime = murder + aggravated assault + robbery.
[c] Property crime = burglary + motor theft + larceny.
[d] The quintile means reported for the final two rows are for the assault crime density quintiles.

means of these averages, again broken down by quintiles, are also reported in Table 3. Once again, there is considerable variation across quintiles, regardless of the measure of crime. In the lower quintiles, the mean changes are generally negative, indicating that, within these neighborhoods, crime tends to fall from one year to the next. In contrast, within the fifth quintile the mean change is always positive, and in most cases this is true within the fourth quintile, as well.

Besides the crime quintiles, at the bottom of Table 3 we report means by quintiles for household income and the rate of house price appreciation, the latter measured over the last four years of the panel, 2004 to 2007.[21] The income quintiles show that the neighborhoods in our sample differ widely in income. In the 20% of neighborhoods with the highest incomes, mean income is more than twice as high as in the 20% of neighborhoods with the lowest incomes. Similarly, there is variation in house price appreciation, with prices rising more than

[21] We compute the rate of house price appreciation over this period because these are the years over which our models attempt to explain changes in home prices. Neighborhood income is obtained by proportioning our neighborhoods among census tracts and then using these proportions to estimate a weighted average.

twice as fast in the top appreciation quintile in comparison to the bottom appreciation quintile.

Finally, instead of using their own quintiles, we report mean household income and the mean appreciation percentage within the aggravated assault crime density quintiles.[22] As expected, average household income is lowest in the top quintile and highest in the bottom quintile. What may appear surprising is that house price appreciation is highest within the highest crime quintile. This seeming anomaly is explained by the tendency of lower priced homes to appreciate at a more rapid rate than higher priced homes within Miami-Dade County over the specified time period (2004–2007).

6. Which measure of crime is best?

In this section we present the results from estimating OLS versions of Eq. (2), using the three alternative measures of crime: the number of crimes, the crime rate and the crime density. These regressions are run to determine which measure performs best in explaining neighborhood housing prices. In addition, we are interested in having OLS estimates of our model that we can compare with those obtained with 2SLS.

Of the three measures, changes in crime density explain the greatest variation in the change in the price index (see Table A.1 in Appendix A). The crime rate, which has been the most used measure of neighborhood crime, outperforms the number of crimes.[23] These results are consistent with those obtained by Bowes and Ihlanfeldt (2001), who also find that crime density outperforms the crime rate in explaining inter-neighborhood variation in housing prices. A number of arguments were made above for why density turns out to be the preferred measure of crime at the neighborhood level. In light of these arguments and the findings presented in Table A.1, we estimate our 2SLS models using crime density as our measure of neighborhood crime.[24]

The OLS crime density results (Column 3 of Table A.1) show that only aggravated assault and robbery crimes have a negative effect on house price. Vandalism is found to have a positive effect.

7. Instrumental variables

Our instrumental variables are lagged changes in the total interior usable square footage of 26 individual commercial land use categories. To determine the optimal structure of the instrument set, we experimented with a number of different lag structures. A combination of changes lagged two and three years (ΔZ_{t-2}, ΔZ_{t-3}) were found to yield the highest joint *F*-statistics in the first-stage regressions of the 2SLS models. These 52 variables form our "full" set of instrumental variables. Our alternative set of instrumental variables purges from the original set those land use categories that identified shopping, eating, and drinking establishments. This reduced the number of land use categories from 26 to 18 and the number of instrumental variables from 52 to 36. The pared down instrument set is referred to as the "reduced" set below.

8. Results from estimating the house price index models

Columns (1) and (2) of Table 4 present the results from estimating Eq. (2) under the assumption that there is no contemporaneous feedback from ΔI_t to ΔC_t (i.e., ΔC_t is instrumented for all seven crimes). The results in Column (1) are obtained using the full set of instruments, while those in Column (2) use the reduced set. In addition to the estimated coefficients (and standard errors robust to heteroskedasticity and serial correlation), Table 4 provides four additional pieces of information: the long run propensity of crime (LRP), the elasticity, Hansen's *J* statistic, and the first-stage *F*-statistics.

The LRP is specific to each type of crime and is obtained by summing up the estimated coefficients on those change variables (e.g., ΔC_{t-3} plus ΔC_{t-4}) that are statistically significant.[25] The LRP represents the change in the house price index that would occur within the neighborhood if there was a permanent increase in the crime measure. Given the LRP, the elasticity is calculated at the point of means.

Hansen's *J* statistic and the first-stage *F*-statistics test the validity of our instrumental variables (IVs). The former is a test of the joint hypothesis that the IVs are uncorrelated with the error term and correctly excluded from the house price index equation. A low *J* statistic value (high *P*-value) allows us to reject the null hypothesis that the IVs are invalid. The first-stage *F*-statistics show whether the IVs are jointly significant in explaining each endogenous crime variable.

Focusing first on the IV tests, regardless of whether the full or reduced IV set is employed, the *J* statistic is very small, providing some confidence that we are using our IVs correctly. The first-stage *F*-statistics are all highly significant, again regardless of the instrument set employed, indicating a close association between the IVs and each of the endogenous crime variables.

The estimated crime coefficients, which are highly consistent between the two alternative sets of IVs, show that only two of the seven types of crime matter to housing values: aggravated assault and robbery.[26] Of the aggravated assault variables, the t and $t-1$ change variables are not significant, but the $t-2$, $t-3$, and $t-4$ changes are significant. The timing pattern revealed by these results suggests, for example, that an increase in the density of assault crime in 2004 (above that reported in 2003) would first lower neighborhood housing prices in 2006 (relative to their values in 2005). For robbery crime, the t, $t-1$, and $t-2$ change variables are insignificant, but the $t-3$ and $t-4$ changes are significant.[27]

The lagged effects that changes in crime have on housing prices carry both methodological and policy importance. Regarding the former, no previous study has addressed the timing issue. The implicit assumption has been that crime has an immediate effect on property values. Our results suggest that this assumption should be added to our list of limitations of previous studies identified in Section 2. Regarding anti-crime policies, it will take time for those policies that have an impact to actually reduce the level of crime in the neighborhood. Our results suggest that even more time will pass before the reduction in crime affects homebuyers' perceptions of the safety of the neighborhood. Combining the crime impact and the perception lags together suggests that a long time frame may be needed to evaluate the effectiveness of particular anti-crime policies in turning around declining neighborhoods.

LRPs for the aggravated assault crime variables equal −1222 and −1311 using the full and reduced set of IVs, respectively. These LRPs, which are both highly significant, translate into elasticities equal to −.152 and −.163. The LRPs for robbery crime are −1485 and −1399

[22] We use the aggravated assault crime quintiles because we find, as reported below, that this type of crime has the strongest effect on property values.

[23] Recall, however, that the crime rate is measured with some error because neighborhood population is only known for the year 2000.

[24] We also use crime density so that we can compare our results to those of Gibbons (2004), who we said above is the only previous study to properly instrument his crime variables (which are also measured in densities).

[25] Statistical significance is specified at the 10% level by a two-tailed test.

[26] The $t-2$ change in the density of murder is positive and significant at the 10% level, but only in Column (1).

[27] Note that these results are not too dissimilar from the OLS results. The differences are that ΔC_t is negative and significant for assault crime and ΔC_t, ΔC_{t-1}, and ΔC_{t-2} are positive and significant for vandalism crime in the OLS model. As noted by a referee, the most important implication that can be drawn from comparing the OLS and 2SLS results is that even after first differencing the data there appears to be an endogeneity bias in the OLS models causing current changes in crime to be statistically significant. The 2SLS results all show that changes in crime have a lagged effect on property values.

Table 4
Estimates of crime density on house price index.

	(1)[a]	(2)[b]	(3)[c]	(4)[d]
Δ Murder				
t	3773	1455	4929	−2971
	(4325)[e]	(4599)	(4591)	(5051)
$t-1$	4712	2860	8503	9380
	(3516)	(3528)	(6304)	(7507)
$t-2$	5262*	4384	9050**	10,021
	(2944)	(2961)	(4140)	(6221)
$t-3$	4299	4218	5741*	7138*
	(3116)	(3060)	(3059)	(4138)
$t-4$	2358	2661	3634	4442
	(2133)	(2185)	(2645)	(3180)
Δ Robbery				
t	−38	−254	221	15
	(639)	(998)	(774)	(1136)
$t-1$	−275	−387	88	762
	(460)	(698)	(881)	(954)
$t-2$	−432	−469	−315	131
	(412)	(534)	(625)	(757)
$t-3$	−797**	−822**	−873*	−821
	(335)	(415)	(477)	(568)
$t-4$	−689***	−577**	−487	−39
	(251)	(264)	(364)	(363)
Δ Assault				
t	−380	−772	−142	−125
	(573)	(811)	(663)	(891)
$t-1$	−381	−581	−727	−1189
	(265)	(375)	(714)	(737)
$t-2$	−368*	−477*	−509	−764*
	(196)	(268)	(364)	(442)
$t-3$	−518***	−529***	−623***	−722***
	(169)	(175)	(240)	(281)
$t-4$	−336**	−305**	−475**	−389**
	(170)	(159)	(198)	(187)
Δ Burglary				
t	319	496	339	255
	(276)	(460)	(331)	(532)
$t-1$	184	302	476	−90
	(213)	(316)	(341)	(314)
$t-2$	−9	25	171	−282
	(132)	(178)	(241)	(222)
$t-3$	117	118	272	−83
	(100)	(119)	(195)	(189)
$t-4$	91	110	218*	142
	(71)	(92)	(116)	(135)
Δ Motor theft				
t	−397	−330	−539	−28
	(389)	(498)	(502)	(596)
$t-1$	−124	−58	−969*	−150
	(198)	(270)	(539)	(535)
$t-2$	−149	−176	−608**	−199
	(114)	(147)	(266)	(283)
$t-3$	63	10	−201	−54
	(114)	(133)	(191)	(208)
$t-4$	−2	−38	−40	−74
	(97)	(116)	(106)	(137)
Δ Larceny				
t	5	130	66	184
	(108)	(156)	(127)	(129)
$t-1$	−8	15	98	−54
	(44)	(59)	(77)	(169)
$t-2$	55	66	85**	21
	(43)	(54)	(42)	(71)
$t-3$	−11	−5	36	4
	(38)	(43)	(47)	(60)
$t-4$	15	−5	4	−54
	(35)	(36)	(37)	(41)
Δ Vandalism				
t	582	705	118	650
	(506)	(575)	(475)	(516)
$t-1$	570	636	631	991
	(401)	(454)	(552)	(547)
$t-2$	474	504	478	560
	(294)	(310)	(415)	(397)
$t-3$	295	338	353	511**
	(233)	(228)	(277)	(251)

Table 4 (*continued*)

	(1)[a]	(2)[b]	(3)[c]	(4)[d]
Δ Vandalism				
$t-4$	95	146	192	292*
	(133)	(134)	(163)	(159)
LRP, robbery	−1485***	−1399**	−873*	
	(519)	(614)	(477)	
Elasticity, robbery	−.111	−.105	−.065	
LRP, assault	−1222***	−1311***	−1098***	−1876***
	(419)	(485)	(375)	(537)
Elasticity, assault	−.152	−.163	−.136	−.233
LRP, motor theft			−1577**	
			(785)	
Elasticity, motor theft			−.287	
Hansen's J	39.1	22.0	37.2	23.2
	[.756][f]	[.822]	[.552]	[.393]
First-stage F				
Murder				
t	10.48	7.93	10.20	9.09
	[.000]	[.000]	[.000]	[.000]
$t-1$			5.02	1.62
			[.000]	[.032]
Robbery				
t	22.64	8.73	16.46	10.91
	[.000]	[.000]	[.000]	[.000]
$t-1$			11.01	6.46
			[.000]	[.000]
Assault				
t	8.96	5.08	5.46	4.69
	[.000]	[.000]	[.000]	[.000]
$t-1$			12.74	9.11
			[.000]	[.000]
Burglary				
t	11.02	7.86	12.77	6.19
	[.000]	[.000]	[.000]	[.000]
$t-1$			24.71	26.41
			[.000]	[.000]
Motor Theft				
t	10.39	11.49	18.88	9.38
	[.000]	[.000]	[.000]	[.000]
$t-1$			24.10	26.69
			[.000]	[.000]
Larceny				
t	60.65	32.27	75.65	32.56
	[.000]	[.000]	[.000]	[.000]
$t-1$			40.56	13.21
			[.000]	[.000]
Vandalism				
t	15.18	9.78	14.17	11.70
	[.000]	[.000]	[.000]	[.000]
$t-1$			12.49	5.46
			[.000]	[.000]
Observations	502	502	502	502

*, **, *** indicates statistical significance at the 10%, 5%, and 1% levels, respectively.

[a] In Column (1) all crime ΔC_t are instrumented. Full set of instrumental variables is used.

[b] In Column (2) all crime ΔC_t are instrumented. Reduced set of instrumental variables is used.

[c] In Column (3) all crime ΔC_t and ΔC_{t-1} are instrumented. Full set of instrumental variables is used.

[d] In Column (4) all crime ΔC_t and ΔC_{t-1} are instrumented. Reduced set of instrumental variables is used.

[e] Standard errors robust to heteroskedasticity and serial correlation in parentheses.

[f] P-values in brackets.

and again are highly significant. Here the elasticities equal −.111 and −.105.[28]

Columns (3) and (4) of Table 4 present the results from estimating Eq. (2) when we allow for contemporaneous correlation between the

[28] Only two of the six studies that we review in Table 1 report elasticities. Naroff et al. (1980) report an elasticity of house value with respect to the total crime rate equal to −1.67, which is implausibly large. Burnell (1988) reports an elasticity of house value with respect to the property crime rate equal to −0.100, which is in line with the size of the elasticities we estimate for robbery and aggravated assault. However, because Burnell only includes the property crime rate in his models, for the reasons we stated in Section 2, not much confidence can be placed in his results.

error term and crime levels (i.e., ΔC_t and ΔC_{t-1} are now instrumented for all seven crimes). The IV tests indicate that the *J* statistics are again small and all (but one) of the first-stage *F*-statistics are significant at better than the 1% level.[29] However, instrumenting a total of 14 variables clearly stretches what can meaningfully be done with our data, especially when using the reduced set of instruments. These results should therefore be interpreted with an appropriate degree of caution.

When using the full set of IVs (Column (3)) the density of aggravated assault and robbery crime again show significant LRPs, although in each case there is one fewer change variable that is significant. However, the estimated elasticities are comparable to those reported in Column (1) obtained under the assumption of no contemporaneous feedback. The assault elasticity is −.136 (vs. −.152 in Column (1)) and the robbery elasticity is −.065 (vs. −.111 in Column (1)). One difference between Column (1) and Column (3) is that in the latter column the $t-1$ and $t-2$ changes in motor theft density are negative and significant, yielding a negative and significant LRP and an elasticity of −.287.

Column (4) reports the results obtained using the reduced set of IVs. While the importance of assault crime increases (LRP = −1876, elasticity = −.233), none of the robbery changes is statistically significant. The significance of the motor theft changes reported in Column (3) also does not carry over to Column (4). The $t-3$ and $t-4$ changes in vandalism are positive and significant.

While the results reported in Column (4) confirm that aggravated assault crime has the greatest effect on housing value, as explained above, less confidence can be placed in these results in comparison to those presented in Columns (1), (2) and (3). Nevertheless, they do introduce some uncertainty over exactly which types of crime, other than assaults, affect property values.

In light of Gibbons' recent results, our results for vandalism crime merit additional comment. We added vandalism crime to our list of types of crime to investigate because Gibbons (2004) found that between burglary and vandalism crime, only the latter affected neighborhood housing values. He found that a one-tenth standard deviation decrease in the local density of vandalism crime adds 1% to the price of an average Inner London property. Gibbons explains his results by arguing that acts of vandalism "motivate fear of crime in the community and may be taken as signals or symptoms of community instability and neighborhood deterioration in general" (Gibbons, 2004: 441).

However, as we noted above, by omitting other types of crime from his model, Gibbons' results may be biased by correlations between vandalism and other types of crime. For example, we find that the crimes that have the greatest impact on house value are aggravated assault and robbery, each of which has a correlation with vandalism equal to .74 (see Table 2).[30] None of our estimated models suggests that the density of vandalism crime negatively impacts housing values. Of course, besides the differences in our models, there are other differences between Gibbons' and our study that may account for the differences in the results. Most obvious is the fact that Inner London and suburban Miami are very different places, where preferences are likely very different as well.

A possible limitation of the results presented in Table 4 is that they are generated from an equation that includes only year dummies as control variables. Omitted variable bias may result if there are variables, other than the crime variables, that are changing over time within neighborhoods that are both correlated with crime and have their own direct effect on neighborhood housing price. We took three approaches to address this issue. First, we added to the model current and lagged changes in three variables: the total interior square footage of commercial space within the neighborhood, the total interior square footage of industrial space within the neighborhood, and the total interior space of multi-family housing within the neighborhood. The idea behind the inclusion of these variables is that they may register externality and access effects that make the neighborhood more or less attractive to homebuyers over time. Only the LRP for multi-family housing is statistically significant, with a positive sign. Apparently, the multi-family variables do capture changes in the attractiveness of the neighborhood that are not accounted for if only the crime and year variables are included in the model. However, the inclusion of the multi-family variables had little effect on the estimated crime LRPs and elasticities.[31]

Our second approach to mitigating possible omitted variable bias was to include a set of dummy variables representing the ten Census County Divisions (CCD) containing our neighborhoods. CCDs are sub-county units that have stable boundaries and recognizable names. A CCD usually represents one or more communities, trading centers or, in some instances, major land uses. The inclusion of the CCD dummy variables allows each of these areas to have its own trend in housing prices, thereby helping to control for unobservables that may be

Table 5
Estimates of Density of Robbery and Assault Crime on House Price Index.

	(1)[a]	(2)[b]	(3)[c]
Δ in period			
t	375	368	315
	(403)[d]	(403)	(417)
$t-1$	−13	83	−22
	(142)	(414)	(456)
$t-2$	−199***	−181*	221
	(68)	(100)	(352)
$t-3$	−493***	−505***	−550
	(94)	(110)	(350)
$t-4$	−431***	−447***	−671*
	(113)	(132)	(374)
LRP	−1123***	−1133***	−1221*
	(167)	(185)	(651)
Elasticity	−.224	−.226	−.243
Hansen's *J*	58.9	57.2	44.4
	(.238)	(.257)	(.622)
First-stage *F*			
t	165.94	122.21	99.55
	[.000][e]	[.000]	[.000]
$t-1$		76.64	40.80
		[.000]	[.000]
$t-2$			314.97
			[.000]
$t-3$			521.22
			[.000]
$t-4$			585.87
			[.000]
Observations	502	502	502

*, **, *** indicates statistical significance at the 10%, 5%, and 1% levels, respectively.
[a] In Column (1) ΔC_t is instrumented.
[b] In Column (2) ΔC_t, ΔC_{t-1} are instrumented.
[c] In Column (3) ΔC_t, ΔC_{t-1}, ΔC_{t-2}, ΔC_{t-3}, ΔC_{t-4} are instrumented.
[d] Standard errors robust to heteroskedasticity and serial correlation in parentheses.
[e] *P*-values in brackets.

[29] The exception is for the ΔC_{t-1} murder variable in the equation using the reduced set of IVs, where the instruments are only significant at the 5% level.

[30] Gibbons is fully aware of the problem created by excluding a violent crime measure from his hedonic model. He offers some evidence using data at a higher level of spatial aggregation (equivalent to counties in the U.S.) that suggests that the omitted variable problem may not be severe. These data show that changes in crime rates of different types of crimes are not highly correlated. But this evidence has little relevance to his model, since his crime measures are not change (i.e., first differenced) variables.

[31] For example, adding the additional control variables to the model assuming no contemporaneous feedback and using the full set of IVs changed the LRP on assault crime from −1222 to −1221 and left the elasticity unchanged at −.152. The robbery LRP (elasticity) changed from −1465 (−.111) to −1192 (−.089).

changing over time within communities. The results obtained from these models closely match those reported in Table 4.[32]

The third approach we took to investigate the possibility of omitted variable bias involved using our instrumental variables. If the crime variables are correlated with the error term as the result of omitted variables, unbiased estimates can be obtained by instrumenting the crime variables. As noted above, in the specification that assumes contemporaneous feedback, we are instrumenting ΔC_t and ΔC_{t-1} for each type of crime. But omitted variables may bias the estimated coefficients on the other crime variables (ΔC_{t-2}, ΔC_{t-3}, and ΔC_{t-4}) that are not being instrumented.

Instrumenting all changes for all crime types is not possible. However, the most credible results presented in Table 4 suggest that aggravated assault and robbery crime are the crimes that affect house value. We therefore added together assault and robbery crimes to form their combined density and used this as our crime measure. Table 5 reports three sets of results using in all cases the full set of IVs: in Column (1) ΔC_t is instrumented, in Column (2) ΔC_t and ΔC_{t-1} are instrumented, and in Column (3) all changes are instrumented. The diagnostics on the IVs are all excellent. Estimated LRPs and elasticities are highly similar across the three specifications and are consistent with those reported in Table 4.[33]

9. Conclusion

In this paper we have outlined numerous mechanisms suggesting that crime is endogenous to neighborhood housing prices. Despite the widespread knowledge of these mechanisms, the vast majority of studies that have investigated the impact of crime on housing values have treated the measure of crime as an exogenous variable. Of the minority of studies that endogenize crime, each study uses a questionable instrumentation strategy or suffers from other possible specification errors.

In an attempt to deal with the endogeneity of crime and avoid other methodological weaknesses found in previous studies, we utilize a nine-year panel of crime for Miami-Dade County at the neighborhood level. Using state-of-the-art panel data estimation techniques we find that home buyers are willing to pay nontrivial premiums for housing located in neighborhoods with less aggravated assault and robbery crime. Our most reliable estimates suggest the elasticities of house value with respect to aggravated assault crime and robbery crime are −.152 and −.111, respectively.

The unimportance of crimes falling within the property crime category (burglary, motor theft, and larceny) may reflect the fact that such crimes cause far less psychic harm to victims in comparison to violent crime (Cohen et al., 1995).[34] Moreover, self-protection measures are more effective in deterring property crimes. In particular, alarm systems can effectively protect residents from both burglary and auto theft and some types of larceny crimes.[35] In comparison, there are a limited number of options that can offer protection against most acts of violence. The best option may be to self-select peaceful living environments, which our results suggest is an option that many households adopt.

[32] For example, the results of the basic model reported in Column (1) of Table 4 show that robbery and aggravated assault elasticities equal −0.111 and −0.152, respectively. After adding the CCD dummy variables to the model, the elasticities are −0.109 and −0.124, respectively.

[33] In the case where we instrumented all changes (Column (3)), the $t-4$ change is significant, but the $t-3$ change falls just below our specified significance level (10%, two-tailed test). However, given that estimated standard errors are going to be inflated by instrumenting so many variables, we included the $t-3$ change in computing the LRP and elasticity.

[34] The statistical insignificance of murder in our models may simply reflect the fact that these crimes rarely appear in our panel. In addition, the motivation for the murder (e.g., drive-by shootings vs. domestic violence) may be relevant to its effect on property value, and we do not have this information.

[35] Home buyers may also feel that their homeowner's insurance protects them from the monetary losses that may result from property crime. Also, fencing and dogs can be an effective property crime deterrent.

Appendix A

OLS estimates of alternative measures of crime on house price index.

	Crime	Crime rate	Crime density
Δ Murder			
t	1.422	10.524	561
	(2.172)[a]	(15.505)	(1781)
$t-1$	2.483	8.720	1242
	(2.404)	(17.010)	(2023)
$t-2$	3.734	10.715	3857*
	(2.445)	(17.819)	(2181)
$t-3$	3.791	7.047	3154
	(2.503)	(16.399)	(2180)
$t-4$	3.032*	2.275	1554
	(1.709)	(11.783)	(1720)
Δ Robbery			
T	.241	.987	57
	(.262)	(1.700)	(250)
$t-1$	.021	−.816	−175
	(.293)	(1.881)	(290)
$t-2$	−.188	−1.722	−420
	(.338)	(1.709)	(283)
$t-3$	−.864***	−5.053***	−879***
	(.286)	(1.794)	(239)
$t-4$	−.624***	−3.672**	−663***
	(.234)	(1.535)	(195)
Δ Assault			
t	−.398***	−1.944**	−251**
	(.138)	(.902)	(125)
$t-1$	−.387**	−1.936*	−220
	(.166)	(1.069)	(158)
$t-2$	.462***	−2.839**	−323*
	(.178)	(1.067)	(168)
$t-3$	.676***	−3.624***	−454***
	(.166)	(1.181)	(161)
$t-4$	−.519***	−2.634**	−316**
	(.175)	(1.146)	(139)
Δ Burglary			
t	−.008	−.265	−45
	(.063)	(.449)	(63)
$t-1$	−.023	.010	−11
	(.081)	(.621)	(87)
$t-2$	−.145*	−.847	−112
	(.084)	(.676)	(87)
$t-3$	.003	.270	−8
	(.061)	(.421)	(61)
$t-4$	.025	−.071	−41
	(.072)	(.531)	(71)
Δ Motor theft			
t	.168	1.643**	214
	(.105)	(.740)	(91)
$t-1$	.134	1.763**	176
	(.120)	(.778)	(104)
$t-2$	−.030	.762	2
	(.102)	(.668)	(90)
$t-3$	.094	1.060	102
	(.113)	(.726)	(100)
$t-4$	−.018	.256	21
	(.087)	(.607)	(81)
Δ Larceny			
t	−.018	−.119	−5
	(.030)	(.228)	(34)
$t-1$	−.009	−.011	−22
	(.036)	(.275)	(37)
$t-2$	.015	.077	29
	(.030)	(.215)	(30)
$t-3$	−.017	−.126	4
	(.026)	(.207)	(29)
$t-4$	.004	.048	27
	(.024)	(.195)	(30)
Δ Vandalism			
t	.344***	1.913**	364***
	(.101)	(.724)	(120)

(continued on next page)

Appendix A (*continued*)

	Crime	Crime rate	Crime density
Δ Vandalism			
$t-1$	.400***	2.165**	523***
	(.140)	(1.040)	(160)
$t-2$	.223	1.122	397***
	(.150)	(1.023)	(140)
$t-3$	.154	.457	190
	(.131)	(.811)	(122)
t-4	.101	.011	108
	(.123)	(.733)	(103)
R-square	.317	.353	.385
Observations	502	502	502

*, **, *** indicates statistical significance at the 10%, 5%, and 1% levels, respectively.
[a] Standard errors robust to heteroskedasticity and serial correlation in parentheses.

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